

Characterize Nash product rule under strategyproofness and core fairness conditions

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Website #40 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Let

$$f : (\Delta^m)^n \rightarrow \Delta^m$$

be a mechanism on the Leontief domain, where for each agent i ,

$$u_i(q) = \min_{j \in M_i} \frac{q_j}{p_{i,j}}, \quad M_i = \{j : p_{i,j} > 0\}.$$

The question is whether the following implication is true:

If f is continuous, strategyproof, and satisfies core fair share (CFS), must one have

$$f(P) = \text{NASH}(P) \quad \text{for every profile } P = (p_1, \dots, p_n) \in (\Delta^m)^n?$$

Equivalently, does there exist a continuous, strategyproof, CFS mechanism f and a profile P with $f(P) \neq \text{NASH}(P)$?

I do not settle the full implication. I prove several rigorous positive results:

- the implication *does* hold for all two-agent instances ($n = 2$);
- it also holds at every profile whose active alternatives are at most two;
- on some larger profile classes, CFS alone already forces the Nash outcome.

Alternative readings considered

The only ambiguity is the meaning of group-strategyproofness. I adopt the standard weak/strict reading used in the cited characterization: no coalition can misreport so that every coalition member weakly gains and at least one strictly gains. Under the stronger “all strictly gain” reading, all arguments below remain valid.

Related literature found

- Brandt, Greger, Segal-Halevi, Suksompong (2025), *Optimal Budget Aggregation with Star-Shaped Preference Domains* — **PARTIAL/TOOL**. This is the central source. It proves that for Leontief utilities, NASH is the unique continuous mechanism satisfying group-strategyproofness and CFS (their Theorem 3), proves the two-alternative strategyproof characterization (their Theorem 6 and Corollary 1), and explicitly lists the exact “strategyproofness versus group-strategyproofness” question as open. (pubsonline.informs.org)

- Brandt, Greger, Segal-Halevi, Suksompong (2025), *Coordinating Charitable Donations with Leontief Preferences* — **TOOL/BACKGROUND**. This related paper establishes structural properties of NASH for Leontief preferences (group-strategyproofness, equilibrium/decomposition interpretations) that underlie some lemmas imported into the MOOR paper. It does not address the exact SP-versus-GSP question here. (sciencedirect.com)
- de Berg, Freeman, Schmidt-Kraepelin, Utke (2024), *Truthful Budget Aggregation: Beyond Moving-Phantom Mechanisms* — **BACKGROUND**. This shows that in multidimensional budget aggregation the space of continuous truthful mechanisms can be richer than in the one-dimensional case, so a naive “truthfulness should force a unique rule” heuristic is unsafe in dimension > 1 . It does not study Leontief utilities or CFS. (arxiv.org)

I searched specifically for later follow-ups on the exact open question and did not find a source claiming a full resolution through my search.

Proof sketch

There are three independent reductions.

First, when $n = 2$, strategyproofness plus CFS already implies group-strategyproofness: singleton coalitions are covered by strategyproofness, and the only larger coalition is the grand coalition N , whose profitable deviation would be a Pareto improvement over the truthful outcome. But CFS implies efficiency on the Leontief domain, so this is impossible. The known characterization with group-strategyproofness then yields $f = \text{NASH}$.

Second, if a given profile uses only two alternatives, then efficiency forces the outcome to remain on that two-dimensional face of the simplex. After identifying that face with the unit interval, we are back in the one-dimensional single-peaked setting. On that restricted domain, CFS implies proportionality on single-minded profiles, and the two-alternative characterization forces the restricted mechanism to be the uniform phantom rule, which is known to coincide with NASH for two-alternative Leontief preferences.

Third, there are profile classes where CFS alone pins down a unique outcome, hence any CFS mechanism must agree with NASH there. Two such classes are:

- single-minded profiles;
- profiles with pairwise disjoint supports.

These are proved directly from the blocking definition.

So the genuine unresolved regime is the one left over after all these reductions: at least three agents, at least three active alternatives, and overlapping supports.

Main result

Theorem 1. *Let $f : (\Delta^m)^n \rightarrow \Delta^m$ be continuous, strategyproof, and satisfy core fair share for Leontief utilities.*

1. *If $n = 2$, then*

$$f(P) = \text{NASH}(P) \quad \text{for every } P \in (\Delta^m)^2.$$

2. More generally, let

$$S(P) := \bigcup_{i=1}^n M_i$$

be the set of active alternatives at profile P . If $|S(P)| \leq 2$, then

$$f(P) = \text{NASH}(P).$$

3. If P is single-minded, i.e. each p_i is a vertex e_{a_i} of Δ^m , then every CFS outcome is uniquely determined and equals $\text{NASH}(P)$.

4. If the supports M_i are pairwise disjoint, then every CFS outcome is uniquely determined and equals

$$\frac{1}{n} \sum_{i=1}^n p_i = \text{NASH}(P).$$

Proof

I will use the following elementary lemma repeatedly.

Lemma 1. *If q satisfies core fair share at profile P , then for every agent i ,*

$$u_i(q) \geq \frac{1}{n}.$$

Equivalently,

$$q_j \geq \frac{p_{i,j}}{n} \quad \text{for every } j \in M_i.$$

Proof. Fix i . If $u_i(q) < 1/n$, let the singleton coalition $G = \{i\}$ choose $q' = p_i$. For any $q'' \in \Delta^m$,

$$\frac{1}{n}q' + \left(1 - \frac{1}{n}\right)q''$$

has j -th coordinate at least $p_{i,j}/n$ for every $j \in M_i$. Hence agent i 's utility from this mixture is at least $1/n$, which is strictly larger than $u_i(q)$. So $\{i\}$ blocks q , contradicting CFS. Therefore $u_i(q) \geq 1/n$. Since

$$u_i(q) = \min_{j \in M_i} \frac{q_j}{p_{i,j}},$$

this is equivalent to $q_j \geq p_{i,j}/n$ for all $j \in M_i$. □

Lemma 2. *If P is single-minded, with $p_i = e_{a_i}$, and $k_j := |\{i : a_i = j\}|$, then every CFS outcome q satisfies*

$$q_j = \frac{k_j}{n} \quad \text{for all } j.$$

In particular, every CFS mechanism agrees with NASH on single-minded profiles.

Proof. Fix an alternative j , and let

$$G_j := \{i : p_i = e_j\}, \quad |G_j| = k_j.$$

Suppose $q_j < k_j/n$. Let coalition G_j choose $q' = e_j$. For any $q'' \in \Delta^m$, each member $i \in G_j$ has utility

$$u_i\left(\frac{k_j}{n}e_j + \left(1 - \frac{k_j}{n}\right)q''\right) = \frac{k_j}{n} + \left(1 - \frac{k_j}{n}\right)q''_j \geq \frac{k_j}{n} > q_j = u_i(q).$$

So G_j blocks q , contradiction. Hence $q_j \geq k_j/n$ for all j . Summing over j ,

$$1 = \sum_j q_j \geq \sum_j \frac{k_j}{n} = 1,$$

so equality must hold throughout; therefore $q_j = k_j/n$ for every j .

Now compute NASH on this profile. Since $u_i(q) = q_{a_i}$,

$$\prod_{i=1}^n u_i(q) = \prod_{j=1}^m q_j^{k_j}.$$

Weighted AM-GM (or Lagrange multipliers for $\sum_j q_j = 1$) shows that the unique maximizer is $q_j = k_j/n$. Thus the unique CFS outcome equals $\text{NASH}(P)$. $\square$

Lemma 3. *If the supports $M_1, \dots, M_n$ are pairwise disjoint, then every CFS outcome equals*

$$\bar{q} := \frac{1}{n} \sum_{i=1}^n p_i.$$

Moreover, $\bar{q} = \text{NASH}(P)$.

Proof. Let q satisfy CFS. By the first lemma,

$$q_j \geq \frac{p_{i,j}}{n} \quad \text{for each } i \text{ and each } j \in M_i.$$

Because the supports are pairwise disjoint,

$$1 = \sum_{j=1}^m q_j \geq \sum_{i=1}^n \sum_{j \in M_i} \frac{p_{i,j}}{n} = \frac{1}{n} \sum_{i=1}^n \sum_{j \in M_i} p_{i,j} = \frac{1}{n} \sum_{i=1}^n 1 = 1.$$

Hence equality holds. Since all summands are nonnegative, we must have

$$q_j = \frac{p_{i,j}}{n} \quad (j \in M_i), \quad q_j = 0 \quad (j \notin \cup_i M_i).$$

Thus $q = \bar{q}$, so the CFS outcome is unique.

It remains to show $\bar{q} = \text{NASH}(P)$. Let $x \in \Delta^m$ be arbitrary, and define

$$s_i := \sum_{j \in M_i} x_j.$$

Then

$$u_i(x) = \min_{j \in M_i} \frac{x_j}{p_{i,j}} \leq \sum_{j \in M_i} p_{i,j} \frac{x_j}{p_{i,j}} = \sum_{j \in M_i} x_j = s_i.$$

Because the supports are disjoint,

$$\sum_{i=1}^n s_i \leq 1.$$

Therefore,

$$\prod_{i=1}^n u_i(x) \leq \prod_{i=1}^n s_i \leq \left(\frac{1}{n} \sum_{i=1}^n s_i \right)^n \leq \left(\frac{1}{n} \right)^n.$$

For $x = \bar{q}$, each $j \in M_i$ satisfies $\bar{q}_j = p_{i,j}/n$, so

$$u_i(\bar{q}) = \frac{1}{n}$$

for every i , and hence

$$\prod_{i=1}^n u_i(\bar{q}) = \left(\frac{1}{n} \right)^n.$$

So $\bar{q}$ attains the maximal Nash product; hence $\bar{q} = \text{NASH}(P)$. $\square$

Lemma 4. *If $n = 2$, then every continuous, strategyproof, CFS mechanism coincides with NASH on all profiles.*

Proof. Let f be continuous, strategyproof, and CFS, and fix a profile P .

By Invoked Result 1, CFS implies efficiency on the Leontief domain. Thus f is efficient.

I claim that f is group-strategyproof. For singleton coalitions this is exactly ordinary strategyproofness. The only other coalition is $N = \{1, 2\}$. Suppose, for contradiction, that there exists a joint misreport P' such that, at the true profile P , both agents weakly prefer $f(P')$ to $f(P)$ and at least one strictly prefers $f(P')$. Then $f(P')$ Pareto-dominates $f(P)$ at profile P , contradicting efficiency of $f(P)$.

So f is continuous, group-strategyproof, and CFS. By Invoked Result 3, $f(P) = \text{NASH}(P)$. As P was arbitrary, $f = \text{NASH}$ on all two-agent profiles. $\square$

Lemma 5. *If $|S(P)| \leq 2$, then $f(P) = \text{NASH}(P)$.*

Proof. Let $S := S(P) = \bigcup_i M_i$.

If $|S| = 1$, then every agent's peak is the same vertex e_j , hence by the single-minded lemma the unique CFS outcome is e_j , which is also $\text{NASH}(P)$.

Now assume $|S| = 2$, say $S = \{a, b\}$. Consider the face

$$\Delta^S := \{q \in \Delta^m : \text{supp}(q) \subseteq S\}.$$

I first show that for every profile $P' \in (\Delta^S)^n$, one has $f(P') \in \Delta^S$. Indeed, by Invoked Result 1, $f(P')$ is efficient; by Invoked Result 2, every alternative receiving positive mass under an efficient Leontief outcome must be critical for some agent. But no agent in P' assigns positive weight to an alternative outside S , so no such alternative can be critical. Hence $f(P')_j = 0$ for every $j \notin S$.

Thus the restriction

$$f^S : (\Delta^S)^n \rightarrow \Delta^S, \quad f^S(P') := f(P'),$$

is well-defined. It is continuous and strategyproof because these properties are inherited by restriction.

Next, f^S satisfies proportionality on this two-alternative domain. Indeed, on any single-minded profile in $(\Delta^S)^n$, the preceding single-minded lemma shows that the outcome is exactly the empirical distribution of peaks, which is the proportional outcome.

Now identify Δ^S with $[0, 1]$ in the usual way. Leontief preferences on two alternatives are single-peaked, and the restricted domain contains exactly one utility per peak. Therefore Invoked Result

4 applies to f^S : f^S must be the uniform phantom rule. By Invoked Result 5, on the two-alternative Leontief domain the uniform phantom rule is exactly NASH. Hence

$$f(P) = f^S(P) = \text{NASH}(P).$$

□

The theorem now follows by combining the last four lemmas. □

Gap to the full statement

What remains unproved is the full characterization for genuinely multidimensional, overlapping-support profiles:

$$n \geq 3, \quad |S(P)| \geq 3,$$

with no special structure such as single-mindedness or disjoint supports.

This is also the regime isolated by the cited MOOR paper: it proves the Leontief characterization with group-strategyproofness, proves the strategyproof version only for the Leximin–Leontief refinement, and explicitly asks whether Theorem 3 still holds under ordinary strategyproofness. (pubsonline.informs.org)

Self red-team

Counterexample attempts.

- I tried to build a counterexample by perturbing NASH inside the CFS-feasible set around the paper’s $n = m = 2$ example where CFS leaves slack. This indeed preserves CFS, but every naive skew I tested creates an immediate profitable deviation for one side or the other.
- I tested natural alternative rules on tiny examples: midpoint/average-peak, max-min utility, and weighted-Nash-style deformations. For instance, with

$$p_1 = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \quad p_2 = (1, 0, 0),$$

the midpoint rule returns $(\frac{3}{4}, \frac{1}{4}, 0)$, but if agent 1 misreports e_2 , the outcome becomes $(\frac{1}{2}, \frac{1}{2}, 0)$, which strictly increases her true utility from $1/2$ to 1 . So these candidates are not strategyproof.

- I also looked for a profile-dependent monotone transformation T with $f = T \circ \text{NASH}$. I found no transformation that visibly preserves all Leontief incentive constraints while also keeping CFS.

Three likely referee objections.

- *Objection 1: In the $n = 2$ argument, why does a beneficial grand-coalition deviation contradict efficiency?* Because under the adopted weak/strict definition of group manipulation, such a deviation produces another feasible outcome that every agent weakly prefers and one strictly prefers at the truthful profile; that is exactly a Pareto improvement.
- *Objection 2: In the two-face reduction, why can’t the restricted mechanism place mass on unused alternatives?* Because CFS implies efficiency (Invoked Result 1), and efficient Leontief outcomes assign positive mass only to alternatives that are critical for some agent (Invoked Result 2). Alternatives outside the face are in no agent’s support, hence cannot be critical.

- *Objection 3: In the disjoint-support lemma, why is $\bar{q} = (1/n) \sum_i p_i$ really the Nash maximizer?* This was checked directly: $u_i(x) \leq s_i := \sum_{j \in M_i} x_j$, the s_i sum to at most 1, and AM-GM yields $\prod_i u_i(x) \leq (1/n)^n$, with equality attained at $\bar{q}$.

Boundary cases checked.

- $n = 1$: trivial, and also covered by the disjoint-support lemma.
- $m = 1$: trivial.
- Single-minded profiles with repeated supporters of the same alternative: the coalition argument yields $q_j = k_j/n$ exactly.
- Profiles on a two-alternative face inside a larger simplex: the efficiency check excludes leakage outside the face.

Ideas tried

1. **Upgrade SP to GSP by efficiency.** This succeeds only when $n = 2$, because then every non-singleton coalition is the grand coalition. *Obstruction in general:* for $n \geq 3$, a two-agent coalition can gain at the expense of outsiders without violating efficiency.
2. **Reduce to a one-dimensional face and use the two-alternative characterization.** This succeeds whenever the active support $S(P)$ has size at most 2. *Obstruction in general:* if $|S(P)| \geq 3$, the restricted problem is genuinely multidimensional and the single-peaked machinery no longer applies.
3. **Replace the group-strategyproof step in Brandt et al.’s Lemma E.4 by continuity plus perturbation.** Idea: if a unilateral misreport is bossy but leaves the manipulator indifferent, perturb the true peak slightly so that the tie breaks and strategyproofness fails. *Obstruction:* ordinary Leontief preferences only inspect the smallest ratio. Changes in higher-ratio coordinates can remain invisible on an entire neighborhood, so the perturbation argument does not automatically break the tie.
4. **Simulate the Leximin–Leontief proof by a sequence of auxiliary reports.** Idea: successively equalize more ratio classes so that ordinary strategyproofness would force equality class-by-class. *Obstruction:* each SP inequality controls only the currently minimal class; unlike Leximin–Leontief, there is no built-in tie-breaker to pin the next class.
5. **Construct a counterexample by perturbing NASH.** I explored convex combinations with average peaks, weighted-Nash deformations, and fixed skews inside the CFS slack region. *Obstruction:* all explicit low-dimensional attempts I checked violate strategyproofness on small examples; I found no viable continuous SP perturbation.

Invoked results

1. **Invoked Result 1 (CFS $\Rightarrow$ efficiency for Leontief utilities).** *Statement.* For Leontief utilities, every core-fair-share outcome is efficient. *Classification.* TOOL. *Citation.* Brandt, Greger, Segal-Halevi, Suksompong (2025), Proposition A.1, *Mathematics of Operations Research*. (pubsonline.informs.org) *Hypothesis audit.* Our utilities are exactly the Leontief utilities of the paper, on the same simplex domain, and our CFS notion matches the one used there.

2. **Invoked Result 2 (efficient Leontief outcomes have only critical positive alternatives).** *Statement.* For Leontief utilities, a distribution q is efficient if and only if every alternative j with $q_j > 0$ is critical for some agent. *Classification.* TOOL. *Citation.* Brandt, Greger, Segal-Halevi, Suksompong (2025), Lemma A.1 (citing Brandt et al. 2025 JET), *Mathematics of Operations Research*. (pubsonline.informs.org) *Hypothesis audit.* Again, the domain is exactly the Leontief simplex domain considered there.
3. **Invoked Result 3 (Leontief characterization under GSP).** *Statement.* With Leontief utilities, NASH is the only continuous mechanism satisfying group-strategyproofness and core fair share. *Classification.* PARTIAL (for the exact question) / TOOL (for my $n = 2$ reduction). *Citation.* Brandt, Greger, Segal-Halevi, Suksompong (2025), Theorem 3, *Mathematics of Operations Research*. (pubsonline.informs.org) *Hypothesis audit.* In the $n = 2$ lemma I prove that f is group-strategyproof; continuity and CFS are given; the domain is the same Leontief simplex domain. Therefore all hypotheses match.
4. **Invoked Result 4 (two alternatives: SP + proportionality $\Rightarrow$ uniform phantom).** *Statement.* For $m = 2$ and an arbitrary domain containing at least one utility for each peak, the only continuous tops-only mechanism satisfying strategyproofness and proportionality is the uniform phantom rule. *Classification.* PARTIAL/TOOL. *Citation.* Brandt, Greger, Segal-Halevi, Suksompong (2025), Theorem 6, *Mathematics of Operations Research*. (pubsonline.informs.org) *Hypothesis audit.* After restricting to a two-alternative face, our domain consists of two-alternative Leontief preferences, which are single-peaked and provide exactly one utility per peak; our mechanisms are tops-only by definition; continuity and strategyproofness are inherited by restriction; proportionality was proved directly on single-minded profiles of the restricted domain.
5. **Invoked Result 5 (two alternatives: uniform phantom = NASH on Leontief).** *Statement.* With two alternatives, NASH for Leontief utilities coincides with the uniform phantom mechanism. *Classification.* PARTIAL/TOOL. *Citation.* Brandt, Greger, Segal-Halevi, Suksompong (2025), Corollary 1, *Mathematics of Operations Research*. (pubsonline.informs.org) *Hypothesis audit.* The restricted profile lies entirely in a two-alternative face, so the corollary applies verbatim after identifying that face with $[0, 1]$.

Self-confidence

3 — I am highly confident in the proved partial results; the only real uncertainty is not in the proofs above but in whether a deeper idea could settle the remaining $n \geq 3$, genuinely multidimensional overlapping-support regime.

Bibliography

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  journal = {Mathematics of Operations Research},
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  author = {Felix Brandt and Matthias Greger and Erel Segal-Halevi and Warut Suksompong},
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  author = {Mark de Berg and Rupert Freeman and Ulrike Schmidt-Kraepelin and Markus Utke},
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